

## Homework 4

Total points : 21

1. Download a multivariate real data set, which is suitable for cluster analysis. Perform  $k$ -means clustering and data depth based clustering along with detailed comparison study.

Points : 7

2. Obtain two real life infinite dimensional data, and check whether they have obtained from the same measure or not.

Points : 7

3. Obtain an infinite dimensional data with two components. Check whether the two components are independent or not.

Points : 7

**Deadline:** Deadline of submission is 10 PM, April 12, 2026.

**Remark:** You can use any package or any programming language to carry out the study.